

Aryan Shah

High School Student
Published Researcher & Engineer

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Profile

Senior in duPont Manual's MST Magnet pursuing genetic, aerospace, and mechanical engineering. Published researcher, international iGEM Gold Medalist, and VEX Robotics Captain with hands-on wet lab and startup experience.

Languages

- English
- Hindi
- Spanish

Technical Skills

- Wet Lab / Genetic Engineering
- Java & Discord Bot Dev
- Figma & Graphic Design
- CAD / 3D Design
- Digital Marketing / Growth
- Forensic Analysis

Standardized Tests

- ACT: 32
- SAT: 1450
- AP Scholar with Honor

Education

2024 – 2027 **duPont Manual High School — MST Magnet** Louisville, KY
Unweighted GPA: 3.48 · Dual Enrollment: Java Programming, Jefferson Community & Technical College (Spring 2025) · Spanish 111, Campbellsville University (Aug 2025 – May 2026) · English 102, University of Louisville (Aug 2026 – May 2027)

Research Experience

Jan 2026 – Present **HS-Kentucky — VP & Promotional Lead** Louisville iGEM
Currently competing in the 2026 iGEM season as part of the HS-Kentucky team.

Nov 2025 – Aug 2026 **Scientific Researcher — Intern** S.Y.A.L.I.S (Remote)
Competitively selected as 1 of 157 researchers from 2,000+ applicants for the 2025 Research Cohort. Conducts multi-disciplinary research under expert mentorship and collaborates with peers globally.

Jan – Dec 2025 **HS-KY Team Member** Louisville iGEM
Executed RNA-Seq experiments and wet-lab protocols investigating PFAS–protein interactions. Co-authored a peer-reviewed publication with the University of Louisville. Designed CAD prototypes, Figma web pages, and outreach graphics for an internationally competing team.

Jan – Nov 2024 **GCM-KY Team Member** Louisville iGEM
Conducted wet-lab experiments, participated in Human Practices outreach, and built V-Cell computational models during the 2024 competition season.

Jun – Nov 2023 **Manual-KY Team Member** Louisville iGEM
Performed wet-lab experiments, researched project information, built V-Cell models, and coordinated fundraising logistics for the team.

Awards & Accomplishments

Oct 2025 **Gold Medal — iGEM Grand Jamboree** Paris, France
Earned international gold medal as part of team HS-KY at the world's premier synthetic biology competition among thousands of global teams.

Mar 2025 **Innovate Award — VEX V5 State Championship** VEX Robotics
Awarded the Innovate Award at the Kentucky VEX V5 High Stakes State Competition (Team 74656A) for innovative robot design and strategy.

Oct 2024 **Gold Medal — iGEM Grand Jamboree** Paris, France
Earned international gold medal as part of team GCM-KY at the world's premier synthetic biology competition among thousands of global teams.

Oct 2023 **Silver Medal — iGEM Grand Jamboree** Paris, France
Earned international silver medal as part of team Manual-KY at the world's premier synthetic biology competition among thousands of global teams.

2024–2026 **Technology Student Association** duPont Manual HS
2nd Place Forensics Regionals (9th) · 3rd Place Mfg. Prototype Regionals (10th) · 3rd Place Forensics State & 2nd Place Regionals (11th).

2024–2026 **Science Olympiad** duPont Manual HS
9th: 3rd Codebusters & Detector Building at State · 10th: 3rd Forensics & 4th Optics at KY Invitational.

2024 (9th) **Quick Recall — Undefeated JV Champions** duPont Manual HS
JV Challenge Champions & JV League Tournament Champions, undefeated.

2025 **AP Scholar with Honor** College Board

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AP Courses & Scores

- AP Comp. Sci. Principles — 3
- AP Seminar — 3
- AP Human Geography — 4
- AP World History — 4
- AP Calculus BC — 3
- AP Physics 1 Algebra Based — 3
- AP English Lang and Comp — 3
- AP US History — 3

Activities & Clubs

- VEX Robotics — Captain / Pres
- Technology Student Assoc. — VP
- iGEM — VP & Promo Lead
- Science Olympiad
- Science Bowl
- Quick Recall
- Creative Science Initiative
- Ultimate Frisbee

Interests

Robotics | Genetic Eng. | Aerospace | Research | 3D Design | Web Design | Marketing | Forensics

Publications

2025 — **Predicting & Validating PFOA Binding to Human Proteins**
Shah, A. et al.
University of Louisville × iGEM Louisville

Oct 2025 — **Systematic Review of Ubiquilin (UBQLN1-4) Expression in Breast Cancer: Prognostic, Mechanistic, and Therapeutic Implications**
Shah, A. et al.
PROSPERO Registered Systematic Review — CRD420251164726

Leadership Experience

2024 – Present — **President & Team Captain** — VEX Robotics, duPont Manual HS
Captain's team 74656S through build and coding; manages club operations and fundraising. Led notebook design for team 74656A, earning the 2025 Kentucky State Innovate Award.

Mar – Aug 2026 — **Discord Developer & Moderator** — Synthica
Built and deployed custom bots, webhook pipelines, and spreadsheet workflows running daily operations for a 1,000+ member global research server. Developed websites with custom content generators.

Apr 2026 – Present — **Associate Journal Editor — Biology** — Synthica
Peer-reviews biology manuscript submissions for the Synthica Journal, delivering structured feedback to student researchers.

Professional Experience

Jan 2026 – Present — **Growth Department Member** — Synthica (Remote)
Drives content strategy and digital marketing; grew Discord community from zero to 1,000+ members in under 14 days.

Mar – May 2026 — **Growth Marketer** — KroniQ AI (Remote)
Produced launch content and social campaigns for an AI productivity platform, driving early user adoption and brand awareness.

Jan – Mar 2026 — **Growth Intern** — Solace (Remote)
Created and distributed content to expand user acquisition and grow the platform's public presence.

Community Service & Outreach

2022 – Present — **Yard Volunteer** — Elderly Neighbors, Louisville
Provides ongoing plant care and watering services for elderly neighbors in the local community.

2022 – Present — **Food Packing Volunteer** — Hindu Temple of Louisville
Regularly packs and distributes meals for community members in need through the temple's outreach program.

2023 – Present — **Promotional Design Lead** — Louisville iGEM
Organizes outreach events and produces digital resources to make synthetic biology accessible to the Louisville community.

2026 – Present — **Research Access Advocate** — Synthica
Connects students worldwide with free mentorship and publication pathways through the Synthica Global Research Challenge.

2021 – 2024 — **Tree Planting Volunteer** — Louisville Grows
Participated in urban reforestation efforts across Louisville over three consecutive years.

2021 – 2024 — **Food Packing Volunteer** — Lifeline Christian Mission
Packed and distributed meals for food-insecure families across the Louisville metro area.